8.3 Volume of Prisms and Pyramids

Name: _______________

Part A: Introduction: In the design of containers and packages, two of the most important measurements to consider are:

__________________ and ______________________________.

The _________________ of a three-dimensional object is a measure of how much space it occupies.

Volume is measured in cubic units, because volume is a three-dimensional measurement.
Typical cubic units for volume are:

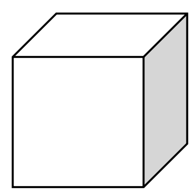
Cubic ____________________ (___)

Cubic ____________________ (___)

Cubic ____________________ (___)

Another unit that is commonly used is the **______** (**__**), which is defined as the volume of a cube with sides of length 10cm.

1 Litre = ______________ cm^3



Litres are generally used when measuring the volume of a liquid. They are often used to describe the capacity of a container.

_____________ is the greatest volume that a container can hold. For example, your family may have a milk jug with a capacity of 1L.

To go from cm^3 to Litres you must: ___________________________________

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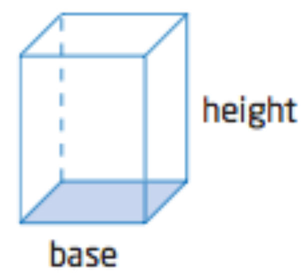
<u>Part B: Volume of Prisms</u>

A prism is a 3-dimensional object with two parallel, congruent polygonal baes. A prism is named by the shape of its base. For example: rectangular prism, and triangular prism.

$$\boldsymbol{Volume\ of\ a\ prism = (Area\ of\ base)(height)}$$

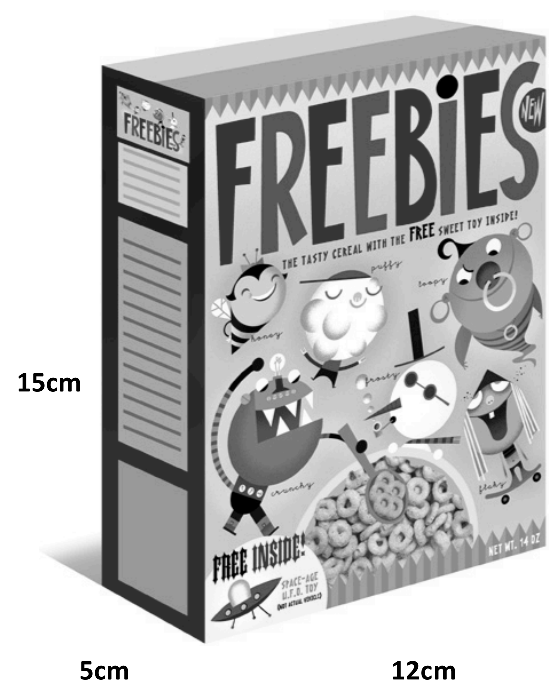


Example 1: Volume of a Rectangular Prism

a) Determine the volume of the package in the photograph, in cubic centimetres.

b) Express the capacity in litres.

Solution:



Example 2: Volume of a Triangular Prism

Determine the volume of the tent.

When finding the volume of a triangular prism you can Use the formula:

$$\boldsymbol{Volume = \frac{1}{2}blh}$$

$b\ stands\ for\ the\ bases\ of\ the\ triangular\ base$
$l\ stands\ for\ the\ height\ of\ the\ triangular\ base$
$h\ stands\ for\ the\ height\ of\ the\ prism$

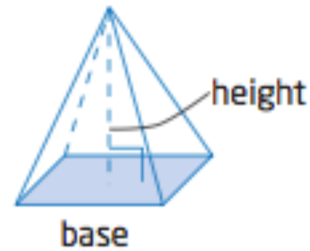
Remember: $\boldsymbol{Volume\ of\ a\ prism = (area\ of\ base)(height)}$

Solution:

Part C: Volume of Pyramids

A pyramid is a polyhedron whose base is a polygon and other faces are triangles that meet at a common vertex.

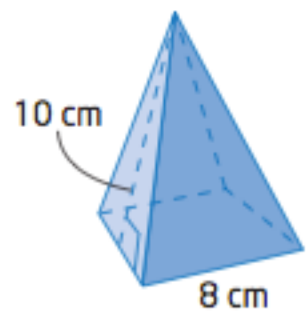
$$\boldsymbol{Volume\ of\ a\ pyramid = \frac{1}{3}(area\ of\ base)(height)}$$



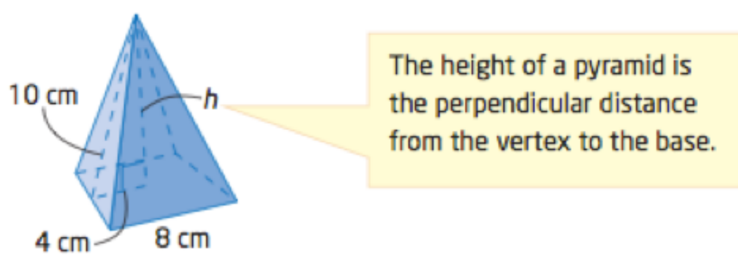
Example 3: Volume of a pyramid

Determine the volume of the square based pyramid-shaped container to the nearest cubic centimeter:

Remember: $Volume = \frac{1}{3}(area\ of\ base)(height)$



First determine the height of the pyramid using **Pythagorean theorem**: $\boldsymbol{a^2 + b^2 = c^2}$



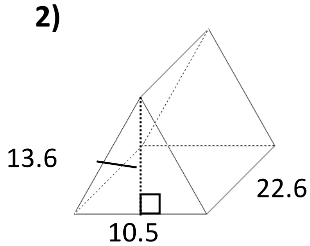
Now that you have the height of the pyramid, you can find the volume:

Part D: Show Me What You've Learned

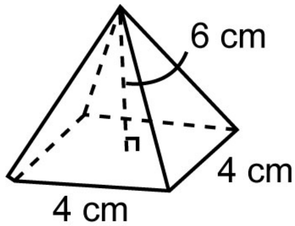
Calculate the volume of the following objects:

1) A cube that has a side length of 5cm:

2)



3)



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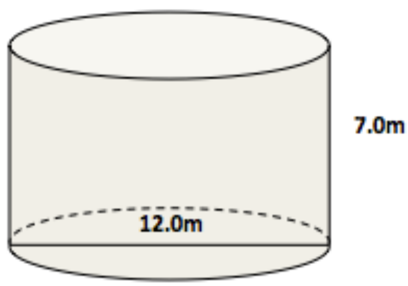

Note: $Volume\ of\ a\ cylinder = \pi r^2 h$

Part E: Conclusion

The volume of a prism is:

The volume of a pyramid is:

Homework: Complete Worksheet